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on

“Design and Implementation of a Dataset-Agnostic, Cost-Aware and Self-Healing
Data Observability Platform”

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CERTIFICATE

Certified that the dissertation entitled *“Design and Implementation of a Dataset-Agnostic, Cost-Aware and Self-Healing Data Observability Platform”* carried out by **Chandrakala J, Rashmi B, Vimal S D, Yuktha S** bearing USN **1MP23IS014, 1MP23IS044, 1MP23IS060, 1MP23IS063** are bonafide students of **BGS College of Engineering and Technology** in partial fulfilment for the award of **“BACHELOR OF ENGINEERING”** in **Information Science and Engineering under VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI** during the academic year **2025 – 26**. It is certified that all corrections indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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DECLARATION BY THE STUDENTS

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Regards,
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ABSTRACT

In modern data-driven systems, data pipelines are essential for analytics and machine learning, yet they often suffer from silent failures, schema changes, data quality issues, uncontrolled cloud costs, and undetected data drift. Most academic solutions remain domain-specific and manually managed, lacking the observability and resilience required for real-world production environments.

This project proposes a dataset-agnostic, cost-aware, drift-aware, and self-healing data observability platform to address these limitations. The system can ingest any structured CSV or JSON dataset without prior schema knowledge, automatically handle schema evolution, dynamically manage warehouse tables, and continuously monitor data quality metrics such as null values, duplicates, freshness, and type inconsistencies. It also applies statistical techniques to detect data drift, ensuring reliability for downstream analytics and machine learning models.

The platform leverages industry-standard tools including Apache Kafka, Apache Airflow, Snowflake, dbt, and Docker to deliver a scalable and production-ready solution. Cloud warehouse usage is monitored to analyze compute consumption and cost trends, while workflow orchestration enables automatic retries and failure recovery. By combining adaptability, observability, and reliability, the system bridges the gap between academic data pipeline projects and real-world data engineering practices.

Vision and Mission of the Institute

Vision:

Creating Competent IT Professionals with Core Values for the Real World.

Mission:

- Providing Students with a Sound Knowledge in IT Fundamentals.
- Exposing Students to Emerging Frontiers in various domains of IT enabling Continuous Learning.
- Promoting Excellence in Teaching, Training, Research and Consultancy.
- Developing Entrepreneurial acumen to venture into Innovative areas of IT.

Imparting value based Professional Education with a sense of Social Responsibility

Vision and Mission of the Department

Vision:

To excel in teaching and innovative research in the field of information science and engineering to support technologically globalized society.

Mission:

- To inculcate strong academic foundation in the information technology domain to empower and equip students for successful career through various teaching learning approaches.
- To identify innovative research-based activities and strengthen entrepreneurial abilities.
- To emphasize the ethical use of technology instilling in our students, a sense of social responsibility towards the betterment of society

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CHAPTER 1

INTRODUCTION

In the era of digital transformation, data has become a critical asset for organizations across diverse domains such as finance, healthcare, e-commerce, education, and scientific research. Modern enterprises increasingly rely on data pipelines to ingest, process, and store large volumes of data from multiple sources in order to support analytics, reporting, and machine learning applications. As data-driven decision-making becomes central to organisational success, the reliability and quality of data pipelines are crucial.

Traditional data pipeline implementations are often tightly coupled to specific datasets and require significant manual intervention to manage schema changes, data quality issues, and pipeline failures. Such systems are prone to silent failures, where incorrect, incomplete, or outdated data propagates to downstream analytics and machine learning models without immediate detection. This can lead to inaccurate insights, reduced model performance, and unreliable business decisions. Additionally, the widespread adoption of cloud-based data warehouses has introduced new challenges related to monitoring and controlling operational costs.

To address these challenges, the concept of data observability has gained importance in modern data engineering practices. Data observability focuses on providing continuous visibility into the health and behavior of data pipelines by monitoring key aspects such as data freshness, volume, distribution, schema evolution, and system reliability. However, most existing academic projects and basic pipeline implementations lack comprehensive observability features, are domain-specific, and do not incorporate mechanisms for cost awareness, data drift detection, or automated recovery.

This project focuses on the design and implementation of a dataset-agnostic, cost-aware, drift-aware, and self-healing data observability platform. The proposed system is capable of handling any structured dataset without prior schema knowledge, automatically adapting to schema changes, continuously monitoring data quality and drift, tracking cloud data warehouse usage costs, and recovering from failures with minimal manual intervention. By leveraging modern

data engineering tools and industry-standard practices, the project aims to provide a robust and production-oriented solution that bridges the gap between academic data pipeline implementations and real-world data engineering systems.

1.1 Significance of the Project

The significance of this project lies in addressing critical challenges associated with modern data pipelines used for analytics and machine learning systems. As organizations increasingly rely on data-driven decision-making, ensuring the reliability, quality, and consistency of data has become a fundamental requirement. This project emphasizes the importance of data observability as a foundational capability for building trustworthy and dependable data systems.

A major significance of the proposed platform is its dataset-agnostic nature. Unlike conventional data pipeline solutions that are tightly coupled to specific datasets or domains, the platform is capable of ingesting and adapting to any structured dataset without prior schema knowledge. This enhances reusability, scalability, and applicability across multiple domains such as finance, healthcare, e-commerce, education, and research.

The integration of data quality monitoring and data drift detection significantly improves the robustness of data pipelines. By continuously tracking metrics such as null values, duplicate records, schema changes, and statistical distribution shifts, the system helps prevent silent data failures and ensures that downstream analytics and machine learning models operate on reliable and stable data. This directly contributes to improved model performance and accurate decision-making.

Another important contribution of this project is its focus on cost awareness. By monitoring cloud data warehouse usage and analyzing compute consumption trends, the platform promotes efficient utilization of cloud resources and helps prevent unnecessary operational costs. This aspect reflects real-world industry concerns, where cost optimization is a critical component of data engineering practices.

The inclusion of self-healing mechanisms further enhances the significance of the project by improving system reliability and resilience. Automatic failure detection, task retries, and recovery reduce downtime and minimize manual intervention, aligning the system with production-grade engineering standards.

Overall, the project is significant in bridging the gap between academic data pipeline implementations and real-world industry systems. By combining observability, adaptability, cost awareness, and resilience into a unified platform, the project provides a strong and practical foundation for modern analytics and machine learning applications.

1.2 PROBLEM STATEMENT

Modern organizations increasingly depend on data pipelines to ingest, process, and store data for analytics, reporting, and machine learning applications. However, many existing data pipeline implementations are domain-specific, tightly coupled to fixed schemas, and require extensive manual intervention to manage changes in data structure and behavior. As a result, these systems are highly vulnerable to silent failures, where data quality issues, schema changes, or abnormal data patterns go undetected and propagate to downstream systems.

In addition, traditional academic and basic industry data pipelines lack comprehensive data observability mechanisms to continuously monitor data quality, data drift, and system reliability. The absence of automated monitoring and alerting leads to delayed detection of errors, unreliable analytics, and degraded machine learning model performance. Furthermore, with the growing adoption of cloud-based data warehouses, uncontrolled compute usage can result in increased operational costs, which are often overlooked in student-level implementations.

There is a clear need for a generalized, intelligent data platform that can operate independently of specific datasets, automatically adapt to schema changes, continuously monitor data quality and drift, track cloud warehouse usage costs, and recover from failures without manual intervention. Addressing these challenges forms the core problem that this project aims to solve.

1.3 RESEARCH MOTIVATION

The motivation for this project arises from the increasing gap between academic data pipeline implementations and real-world data engineering systems used in industry. While academic projects often focus on model building or simple data processing tasks, they rarely address critical production challenges such as data reliability, observability, scalability, and cost management. This gap limits students' exposure to real-world data engineering practices and system-level thinking.

Another key motivation is the growing importance of data observability in modern analytics and machine learning workflows. In real-world deployments, ensuring that data is accurate, consistent, and stable over time is essential for maintaining the performance and trustworthiness of analytical models. The lack of automated mechanisms to detect data quality issues and data drift often leads to incorrect insights and model failures, motivating the need for a robust observability-driven solution.

The project is also motivated by the need to design systems that are adaptable and reusable across multiple domains. By developing a dataset-agnostic platform capable of handling any structured dataset, the project promotes scalability and long-term usability rather than one-time, domain-specific solutions. Additionally, incorporating cost awareness and self-healing mechanisms reflects practical industry concerns, where efficient resource utilization and system reliability are critical.

Overall, the motivation of this research is to design and implement a production-oriented, intelligent data observability platform that not only addresses current limitations in academic projects but also equips students with hands-on experience in modern data engineering architectures and best practices.

CHAPTER 2

LITERATURE REVIEW

In recent years, extensive research has been carried out to improve the reliability, monitoring, and efficiency of data pipelines supporting analytics and machine learning systems. The existing literature addresses various aspects such as self-healing mechanisms, anomaly detection, orchestration efficiency, cloud governance, and data drift detection. A summary of the most relevant studies reviewed for this project is presented below.

In the paper *Self-Healing Data Pipelines: Leveraging AI to Detect and Correct Failures in Real-Time* (2026) [1], Patel and Dey propose an AI-driven framework capable of identifying pipeline failures in real time and triggering automated corrective actions. The study demonstrates how self-healing mechanisms can reduce downtime and manual intervention in data pipelines. However, the work primarily focuses on failure recovery and does not address dataset-agnostic ingestion, dynamic schema evolution, or cost-aware data warehouse monitoring.

In *Proactive Data Pipeline Maintenance via Machine Learning-Driven Anomaly Detection* (2025) [2], Sharma et al. present a machine learning-based approach to proactively detect anomalies in data pipelines before critical failures occur. The proposed system enhances early issue detection but relies on historical training data and predefined features, limiting its adaptability to rapidly changing datasets and schema variations.

The study *Cost-Effective Big Data Orchestration Using Dagster: A Multi-Platform Approach* [3] explores the role of workflow orchestration frameworks in optimizing resource utilization and reducing operational costs across multiple big data platforms. While the paper highlights cost efficiency at the orchestration level, it does not integrate data quality monitoring, data drift detection, or observability at the data warehouse layer.

In *Governing Cloud Data Pipelines with Agentic AI* [4], the authors introduce an agentic AI-based approach for governing and managing cloud data pipelines through autonomous decision-making and policy enforcement. Although the paper presents an advanced governance model, it remains largely conceptual and does not emphasize practical implementation aspects such as dataset-agnostic schema handling or continuous data quality observability.

The paper *Model Monitoring, Data Drift Detection, and Efficient Model Retraining: A Review* [5] provides a comprehensive survey of techniques for monitoring deployed machine learning models and detecting data drift. The study discusses statistical and learning-based drift detection

methods and their role in maintaining model performance. However, the focus is primarily on post-deployment model monitoring rather than integrating drift detection directly into the data pipeline stage.

A Survey on Data Drift Detection Methods in Machine Learning [6] reviews a wide range of data drift detection techniques, including statistical tests and adaptive learning approaches. While the survey offers valuable theoretical insights, it does not address the practical challenges of embedding drift detection into end-to-end data pipelines with automated observability, cost awareness, and self-healing capabilities.

H. Foidl et al. [7], presented a comprehensive study on data pipeline quality by analyzing influencing factors, root causes, and process-related issues in large-scale analytics systems. The authors emphasized challenges such as schema evolution, missing data, and delayed pipeline failures that often go unnoticed. While the study provides a strong conceptual understanding of pipeline quality issues, it does not propose an automated observability or self-healing framework to actively detect and mitigate these issues in real time.

S. Krishnan et al. introduced *ActiveClean* [8], an interactive framework that integrates data cleaning with machine learning model training. The approach focuses on incrementally improving model accuracy by prioritizing corrupted data during training. However, the solution operates at the model-training level and lacks pipeline-level observability, automated failure recovery, and cost-awareness, making it unsuitable for complex, end-to-end production data pipelines.

A. Akidau et al. [9] proposed the Dataflow programming model to balance correctness, latency, and cost in large-scale data processing systems. The model introduces advanced concepts such as event-time processing and windowing for reliable stream analytics. Although the framework improves pipeline correctness, it does not address data quality monitoring, anomaly detection, or self-healing mechanisms required for maintaining long-term pipeline reliability.

J. Gama et al. [10] provided an extensive survey on concept drift adaptation techniques, categorizing drift into sudden, gradual, and recurring types. The study reviews multiple adaptive learning strategies for handling drift in streaming data environments. Despite its strong theoretical foundation, the work does not explore how drift detection can be operationalized within modern data pipelines with automated monitoring, observability dashboards, or remediation workflows.

V. Hodge and J. Austin [11] surveyed a wide range of outlier and anomaly detection techniques applicable to time-series and streaming data. The paper highlights statistical, distance-based, and density-based methods for anomaly identification. However, the survey does not consider the

integration of these methods into data engineering pipelines nor their role in enabling automated root cause analysis or self-healing pipeline behavior.

M. Zaharia et al. [12] introduced Apache Spark as a unified engine for large-scale data processing, significantly improving performance and scalability for batch and streaming workloads. While Spark provides fault tolerance at the execution level, it lacks native support for data observability, quality validation, cost optimization, and intelligent failure recovery across interconnected pipelines.

S. Ramaswamy and P. Kulkarni [13] proposed a set of data quality metrics for large-scale analytics systems, focusing on accuracy, completeness, and consistency. The study highlights the importance of measuring data quality to ensure reliable analytics. However, the work remains metric-centric and does not provide an automated monitoring or alerting mechanism that continuously enforces these metrics within dynamic data pipelines.

J. Lin et al. [14] introduced a symbolic representation technique for time-series data that enables efficient similarity search and anomaly detection in streaming environments. While effective for detecting deviations in data distributions, the approach operates in isolation and does not address how detected anomalies can trigger automated pipeline-level responses or self-healing actions.

A. Subramaniam and K. Patel [15] discussed data reliability engineering practices for modern analytics platforms, emphasizing monitoring, incident response, and reliability metrics. Although the study acknowledges the need for reliable data systems, it relies heavily on manual interventions and does not propose an AI-driven, cost-aware, or autonomous observability framework capable of handling failures at scale.

Overall, the reviewed literature addresses individual challenges such as pipeline failure recovery, anomaly detection, orchestration efficiency, governance, and data drift detection. However, none of the existing works proposes a unified, dataset-agnostic data observability platform that simultaneously integrates dynamic schema handling, continuous data quality monitoring, cost-aware warehouse usage, and self-healing mechanisms. This observation motivates the need for the proposed system.

2.1 RESEARCH GAP

The analysis of the reviewed literature reveals several critical gaps in existing research related to data pipeline reliability, observability, and management. Although individual studies address

specific challenges such as self-healing mechanisms, anomaly detection, orchestration efficiency, governance, and data drift detection, a comprehensive and unified solution is still lacking.

Most existing works on self-healing data pipelines focus primarily on detecting failures and initiating recovery actions. However, these approaches do not support dataset-agnostic data ingestion or dynamic schema evolution, limiting their applicability across diverse datasets and rapidly changing data environments. As a result, manual intervention is still required when data structures evolve.

Research on machine learning-driven anomaly detection for proactive pipeline maintenance relies heavily on historical training data and predefined features. While effective in controlled environments, such methods are less adaptable to dynamic and unseen datasets, making them unsuitable for generalized, reusable data pipeline platforms.

Although orchestration frameworks and cloud governance models emphasize efficient resource utilization, there is limited research that integrates explicit cloud data warehouse cost monitoring as a core observability metric. Cost awareness is often treated as an operational concern rather than an integral component of data pipeline design, especially in academic implementations.

Furthermore, existing studies on data drift detection predominantly focus on model-level monitoring after deployment. Drift detection is rarely integrated directly into the data pipeline layer, where early identification of abnormal data behavior could prevent the propagation of unreliable data to analytics and machine learning models.

Another significant gap is the lack of end-to-end observability in current systems. Data quality monitoring, schema validation, drift detection, and failure recovery are typically implemented as isolated components rather than as part of a unified, automated platform. This fragmented approach reduces system reliability and increases operational complexity.

Therefore, there exists a clear research gap for a dataset-agnostic, observability-driven data platform that integrates dynamic schema handling, continuous data quality monitoring, statistical data drift detection, cost-aware cloud warehouse usage, and self-healing mechanisms within a single unified architecture. Addressing this gap forms the primary focus of the proposed project.

CHAPTER 3

OBJECTIVES OF THE PROJECT

The primary objective of this project is to design and implement a dataset-agnostic, cost-aware, drift-aware, and self-healing data observability platform that ensures reliable, scalable, and production-ready data pipelines for analytics and machine learning applications.

The specific objectives of the project are as follows:

1. To design a dataset-agnostic data ingestion framework capable of accepting any structured dataset in CSV or JSON format without prior schema knowledge.
2. To implement dynamic schema detection and automated table management, enabling the system to adapt seamlessly to schema changes without manual intervention.
3. To develop continuous data quality monitoring mechanisms to track metrics such as null values, duplicate records, data freshness, and data type inconsistencies.
4. To incorporate statistical data drift detection techniques for identifying significant changes in data distributions and abnormal data behavior over time.
5. To enable cost-aware data pipeline monitoring by tracking cloud data warehouse compute usage and analyzing cost consumption trends.
6. To design and implement self-healing mechanisms that automatically detect failures, retry tasks, and recover pipeline execution with minimal manual effort.
7. To build a scalable and modular system architecture using modern data engineering tools that reflects real-world industry practices.

CHAPTER 4

PROPOSED SYSTEM

The proposed system is a dataset-agnostic, cost-aware, drift-aware, and self-healing data observability platform designed to ensure reliable and production-ready data pipelines for analytics and machine learning applications. The system is capable of ingesting any structured dataset in CSV or JSON format without prior schema knowledge and automatically adapting to schema changes.

The platform follows a modular architecture built using modern data engineering tools. Data ingestion is handled through a streaming-based approach, enabling scalable and flexible data intake. Workflow orchestration is managed using an automated scheduler to coordinate ingestion, schema detection, data loading, quality checks, drift detection, and cost monitoring tasks.

Raw data is stored in a cloud-based data warehouse, where dynamic schema detection logic automatically generates and manages structured tables. Data transformation and profiling processes compute essential statistics and metadata required for observability. Continuous monitoring mechanisms are integrated to track data quality metrics, detect data drift, and log warehouse compute usage. In the event of failures, self-healing mechanisms automatically retry tasks and recover pipeline execution without manual intervention.

By integrating observability, adaptability, cost awareness, and resilience into a unified framework, the proposed system provides a robust data foundation for downstream analytics and machine learning systems.

4.1 METHODOLOGY

The project follows a systematic and incremental development methodology to ensure robustness and practical feasibility. Initially, requirements are analyzed to identify key challenges related to data ingestion, schema evolution, data quality, drift detection, cost monitoring, and fault tolerance.

In the next phase, a prototype data pipeline is developed to support dataset-agnostic ingestion and dynamic schema detection. Workflow orchestration is implemented to automate pipeline

execution and manage task dependencies. Data quality checks and statistical profiling techniques are then integrated to monitor the health of incoming data.

Subsequently, statistical data drift detection methods are applied to identify deviations in data distributions over time. Cost-aware monitoring is implemented by tracking cloud data warehouse compute usage and analyzing consumption trends. Finally, self-healing mechanisms are incorporated through automated retries, error logging, and failure recovery strategies.

The entire system is containerized to ensure reproducibility and ease of deployment. Each development phase is validated through testing and iterative refinement to improve system reliability and performance.

4.2 EXPECTED OUTCOMES

Upon successful completion of the project, the proposed system is expected to deliver the following outcomes:

1. A fully functional dataset-agnostic data pipeline capable of handling structured datasets from multiple domains.
2. Automated schema detection and adaptation, reducing manual intervention during data structure changes.
3. Continuous monitoring of data quality metrics, ensuring reliable and consistent data for analytics and machine learning.
4. Early detection of data drift to prevent the propagation of abnormal data behavior to downstream systems.
5. Effective monitoring of cloud data warehouse usage, promoting cost-aware and efficient resource utilization.
6. Improved pipeline reliability through self-healing mechanisms that minimize downtime and operational overhead.
7. A scalable and production-oriented data observability platform that bridges academic learning with real-world data engineering practices.

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